Entergy Operations, Inc.
River Bend Station
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Tel 225-381-4374

William F. Maguire
Site Vice President

RBG-47872

June 19, 2018

Attn: Document Control Desk
U.S. Nuclear Regulatory Commission
11555 Rockville Pike
Rockville, MD 20852-2738

Subject: Licensee Event Report 50-458 / 2018-002-00

River Bend Station, Unit 1
Docket No. 50-458
License No. NPF-47

Dear Sir or Madam:

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Sincerely,

WFMaguire

WFM/twf

Enclosure

cc: (with Enclosure)

U.S. Nuclear Regulatory Commission
Attn: Ms. Lisa M. Regner, Project Manager
09-D-14
One White Flint North
11555 Rockville Pike
Rockville, MD 20852

U.S. Nuclear Regulatory Commission
Region IV
1600 East Lamar Blvd.
Arlington, TX 76011-4511

NRC Senior Resident Inspector
Attn: Mr. Jeff Sowa
5485 U.S. Highway 61, Suite 1
St. Francisville, LA 70775

Public Utility Commission of Texas
Attn: PUC Filing Clerk
1701 N. Congress Avenue
P. O. Box 13326
Austin, TX 78711-3326

INPO
(via ICES reporting)

RBF1-18-0116

NRC FORM 366 (04-2018)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
River Bend Station - Unit 1	05000 458	1 OF 4

4. Title

Inadvertent High Pressure Core Spray Initiation and Loss of Safety Function due to Inadequate Work Instruction Mitigation Actions.

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name NA	Docket Number 05000 NA
04	26	2018	2018 -	002 -	00	06	19	2018	Facility Name NA	Docket Number 05000 NA

9. Operating Mode: 1

10. Power Level: 100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☑ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Tim Schenk, Manager - Regulatory Assurance	225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date)	☑ No		NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

At 15:31 CST on April 26, 2018, I&C technicians were restoring Reactor Vessel Water Level Transmitter B21-LTN081C to service with the Reactor operating in Mode 1 at 100% power. During this transmitter restoration, the station experienced an invalid initiation of High Pressure Core Spray (HPCS) with injection into the reactor vessel. The HPCS Diesel Generator also started per design, but did not connect to the switchgear because there was no bus under voltage. Operations terminated the injection after approximately 40 seconds in accordance with Alarm Response Procedures. The Feedwater Level Control System responded per design, and maintained nominal Reactor water levels. The cause of this event was inadequate work instruction. This event was of minimal significance to the health and safety of the public.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
River Bend Station - Unit 1	05000- 458	2018	002	00

NARRATIVE

BACKGROUND

High Pressure Core Spray (HPCS) [BG] is designed to limit the release of radioactive materials to the environment following a loss of coolant accident (LOCA) by injecting water into the Reactor Coolant System (RCS). HPCS is also designed to automatically initiate when level 2 is reached in the Reactor Pressure Vessel (RPV). The RPV level 2 signal is generated by Reactor Water Level Transmitters (**LT**) in the Nuclear Boiler Instrumentation System. The 'C' reference leg of the Nuclear Boiler Instrumentation System contains two HPCS level 2 Water Level Transmitters.

REPORTED CONDITION

On April 26, 2018 after replacing Reactor Water Level Transmitter B21-LTN081C, an invalid initiation occurred resulting in the injection of HPCS.

With the Reactor operating in Mode 1 at 100% power, I&C technicians were sent to the field to replace the transmitter which is in the 'C' reference leg. In the process of returning the transmitter to service, air introduced during the installation caused a perturbation in the 'C' reference leg. With two HPCS level transmitter on the 'C' reference leg affected by this perturbation, HPCS initiated and the Division III Diesel Generator started as designed. Operators responded to the initiation by taking the appropriate actions per alarm response procedure. The HPCS system was placed in standby, the plant was stabilized and field activities stopped. All plant parameters responded as would be expected for an inadvertent actuation of the HPCS System. NRC notification of Emergency Core Cooling System (ECCS) discharge to RCS and subsequent disabling of a single train system (HPCS) was completed at 18:50 CDT on April 26, 2018 (EN 53365).

A troubleshooting procedure was developed and utilized to restore 'C' reference leg following the event.

Although initially reported in accordance with 10CFR50.72(b)(2)(iv)(A), it has since been determined that the actuation signal to the HPCS system was invalid. This report is submitted in accordance with 10CFR50.73(a)(2)(iv)(A) for the invalid actuation of HPCS. This report is also submitted in accordance with 10CFR50.73(a)(2)(v)(D) because the HPCS system was placed in standby and could not fulfill its safety function for a short period following this event.

CAUSAL ANALYSIS

An organizational false sense of security existed based on past successful on line completion of a Reactor Water Level Surveillance Test Procedure (STP). The mitigation strategy of this STP was used in the transmitter replacement work package. This faulted mental model influenced inadequate risk recognition during the analysis, design, development, and implementation of solutions. As a result, the risk screening conducted was not adequate for the work performed. Work order instructions were not written in sufficient detail to preclude the event.

CORRECTIVE ACTION TO PREVENT RECURRENCE

The following actions have been assigned to prevent a recurrence of this event and are documented in the station's corrective action program.

NARRATIVE

- Conduct a review of Surveillance Test Procedures that impact reference leg instrumentation with ECCS initiation function.

- Production Manager set and document expectations for work package detail and process rigor with Planning Department staff in accordance with station and corporate procedures.

- Online Maintenance Scheduling Superintendent set and document expectations for process rigor with Work Control staff.

- Engineering Managers set and document expectations for process rigor for all technical activities.

- Maintenance Manager set and document expectations for process rigor with Maintenance staff.

- Develop and present a Case Study to all employees in the applicable departments focusing on the Organizational shortfalls that led to injection of HPCS.

PREVIOUS OCCURRENCE EVALUATION

No similar events have been reported by River Bend Station in the last three years.

SAFETY SIGNIFICANCE

The RBS Updated Safety Analysis Report addresses an inadvertent initiation of HPCS. The addition of cooler water to the upper plenum causes a reduction in steam flow which results in some depressurization as the pressure regulator (**RG**) responds to the event. In the automatic flow control mode, following a momentary decrease, neutron power settles out at a level slightly above operating level. In manual mode, the flux level settles out slightly below operating level. In either case, pressure and thermal variations are relatively small and no significant consequences are experienced. The minimum critical power ratio remains within the safety limit and, therefore, fuel thermal margins are maintained.

The ECCS is designed, in conjunction with the primary and secondary containment, to limit the release of radioactive materials to the environment following a LOCA. The ECCS uses two independent methods (flooding and spraying) to cool the core during a LOCA. The ECCS network is composed of the HPCS System, the Low Pressure Core Spray (LPCS) System, and the low pressure coolant injection (LPCI) mode of the Residual Heat Removal (RHR) System. The ECCS also consists of the Automatic Depressurization System (ADS). The suppression pool provides the required source of water for the ECCS. Although no credit is taken in the safety analyses for the condensate storage tank (CST), it is capable of providing a source of water for the HPCS System.

Adequate core cooling is ensured by the operability of the redundant and diverse low pressure ECCS injection/spray subsystems in conjunction with the ADS. Also, the Reactor Core Isolation Cooling (RCIC) System will automatically provide System will automatically provide makeup water at most reactor operating pressures. LPCS, LPCI, ADS, and RCIC were operable for the duration of the event.

This event was of minimal significance to the health and safety of the public due to the availability of alternate ECCS and the negligible effect of an inadvertent initiation of HPCS on reactor safety.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

(NOTE: Energy Industry Identification System component function identifier and system name of each component or system referred to in the LER are annotated as (**XX**) and [XX], respectively.)